Broker 4: Summaries and Outcomes

Norah Kingsland
301580336

Current game vs modified version with clear signifier

For our Broker 1 assignment focusing on colour, my team and I focused on Pokémon GO, a mobile game in which users find and catch virtual Pokémon in an Augmented Reality version of the real world. Using the game in a black and white monochrome filter revealed three key areas of the game that used exclusively colour to convey information: Gyms/Pokestops, the Pokémon catch ring when catching a Pokémon, and the man you use to navigate the game. We set out to improve the game's accessibility through higher contrast and clearer signifiers, testing it with colour blind filters.

Easier to see the difficulty of catch

Broker 1: Colour

Broker 2: Signifiers and Affordances

Broker 3: Distributed Cognition

New proposed accessibility settings

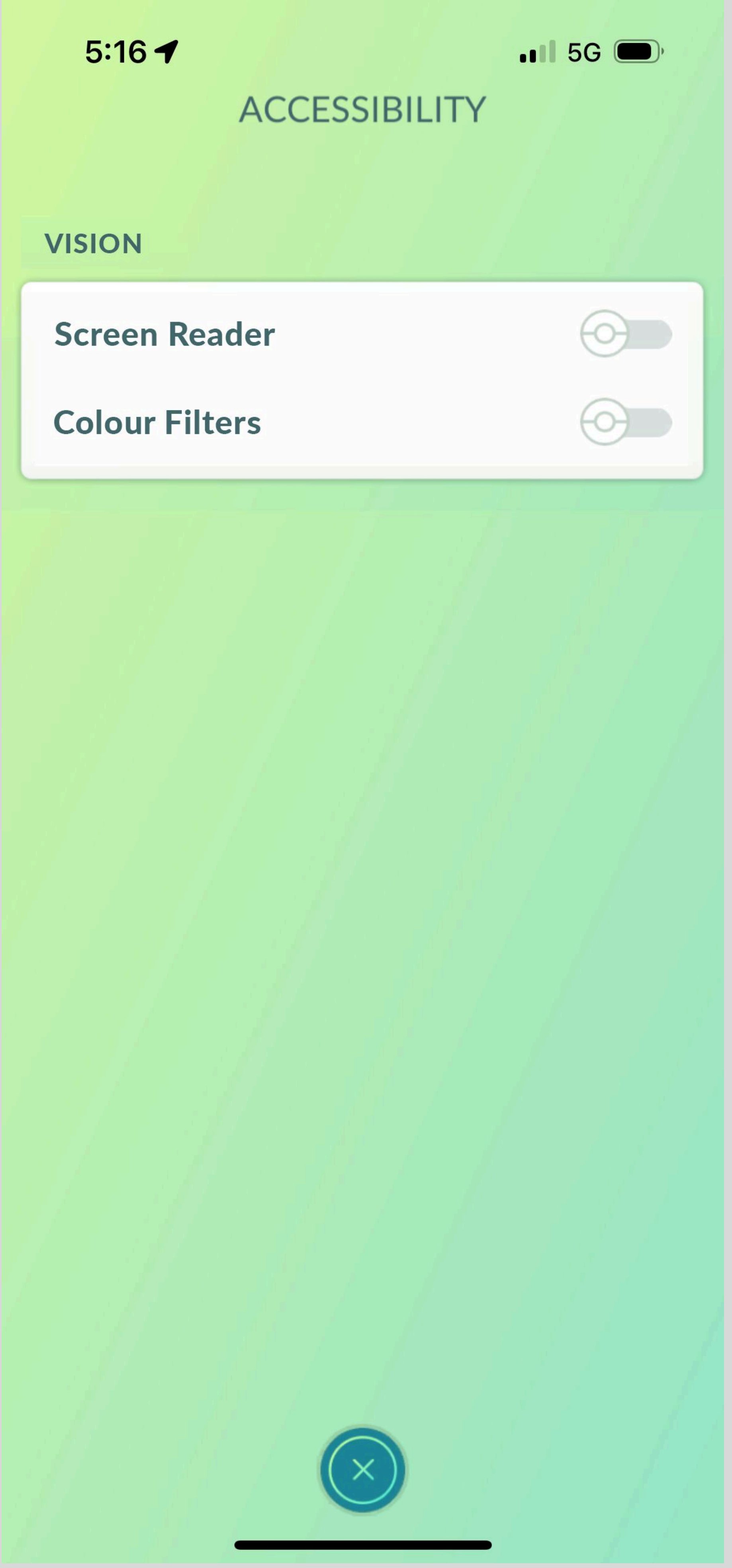
Categories (such as vision) are grouped using frames to show relatedness. In the future, other categories (such as hearing for audio impairment) could be added.

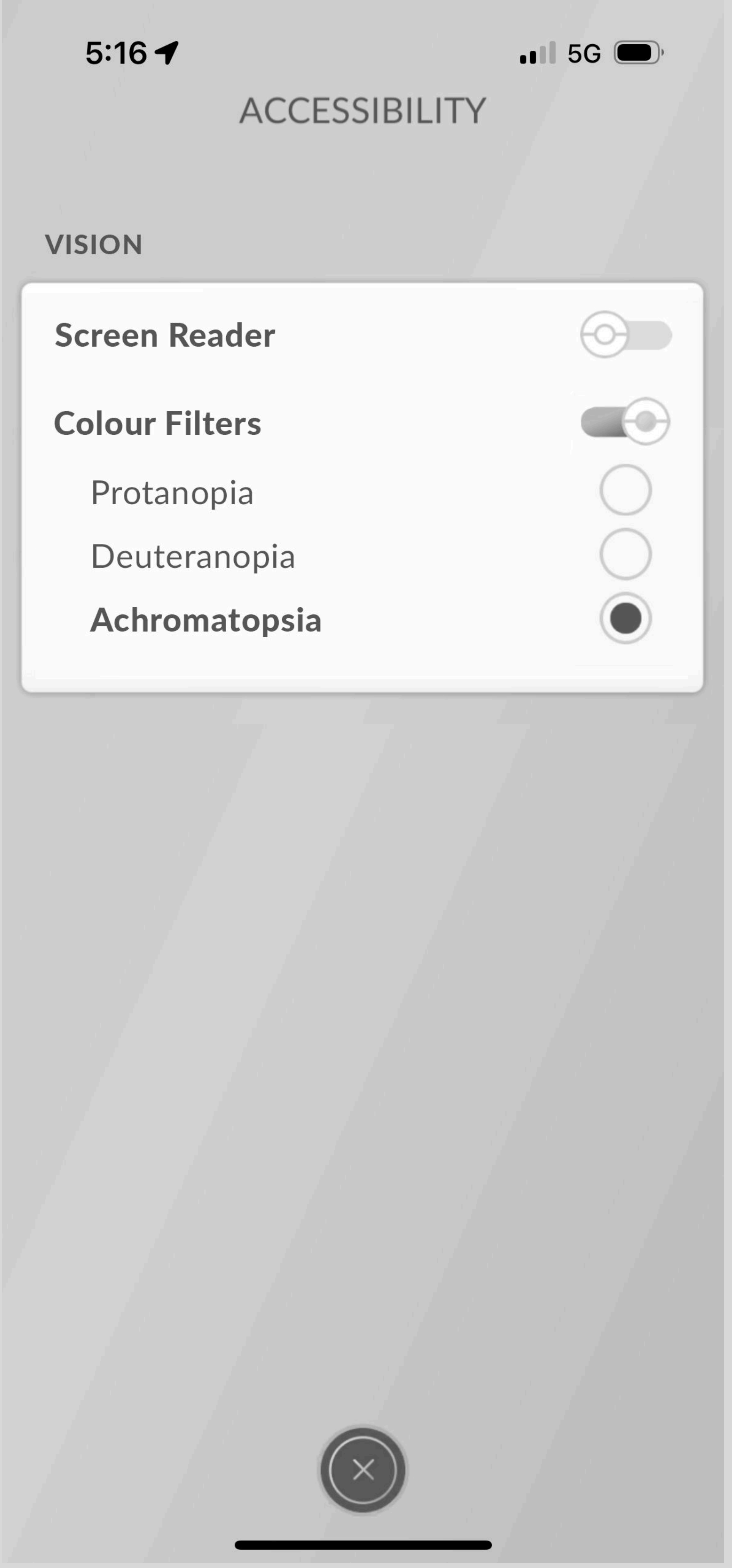
A key thing I took from this project was the idea of designing with colour blindness, as well as other potential user struggles, in mind. While I have always checked colour contrasts for the overall visibility of my designs, it had never occurred to me to check things in gray scale or other colour blind filters.

My group's research process was essential to find strategies to make things more accessible, such as ensuring high contrast colours, avoiding using colour combinations such as red/green, green/gray, etc, and avoiding showing information solely through colour. Our research also encouraged us to check the design in a variety of colour blind filters and get feedback from affected individuals, which are great ways to use the iterative designing process and demonstrate evidence based design.

These designs pushed me to challenge my relationship with colour in design. I often default to using colour as a signifier, and trying out new methods of conveying information, such as meaningful icons, symbols, textures, or patterns that are more accessible was great to advance my skills as an evidence based designer.

We applied concepts such as smart seeing in our overhaul of the app's Accessibility Settings, pictured to the right, where we changed the interface to group settings by the accessibility concern they address, utilizing smart seeing and the gestalt principle of proximity to help the user use it intuitively.





Gym Types, colours altered for achromatopsia (Broker 1)

Contrast increased, patterning added for more visual clarity

My team and I primarily resolved these colour difficulties by increasing the contrast of the colours.

Reflecting on the assignment, we did use signifiers in some scenarios (such as the “Easy” icon in the catching menu) to increase the apps clarity for colourblind users. However, after learning more about affordances and signifiers, in the future I would focus much more on using signifiers to make the games interactions more clear rather than just increasing contrast. Contrast remains an important and useful tool, but it may not be sufficient on its own.

For the gym types especially, something that relies on more than just the broken apart small coloured ring at the top to show which team the gym is would be the first place I’d start. Currently, even with the increased contrast, the size of the signifier may be hard to see for those with vision impairment

In the future: the icon on top of gyms could be changed to icons associated with Team Valor, Mystic, or Instinct, the game’s 3 gym types. This makes the game more accessible and implements meaningful signifiers, rather than the current one, which just tells the user “I am an unclaimed gym.” These icons should also have more contrast with their surroundings.

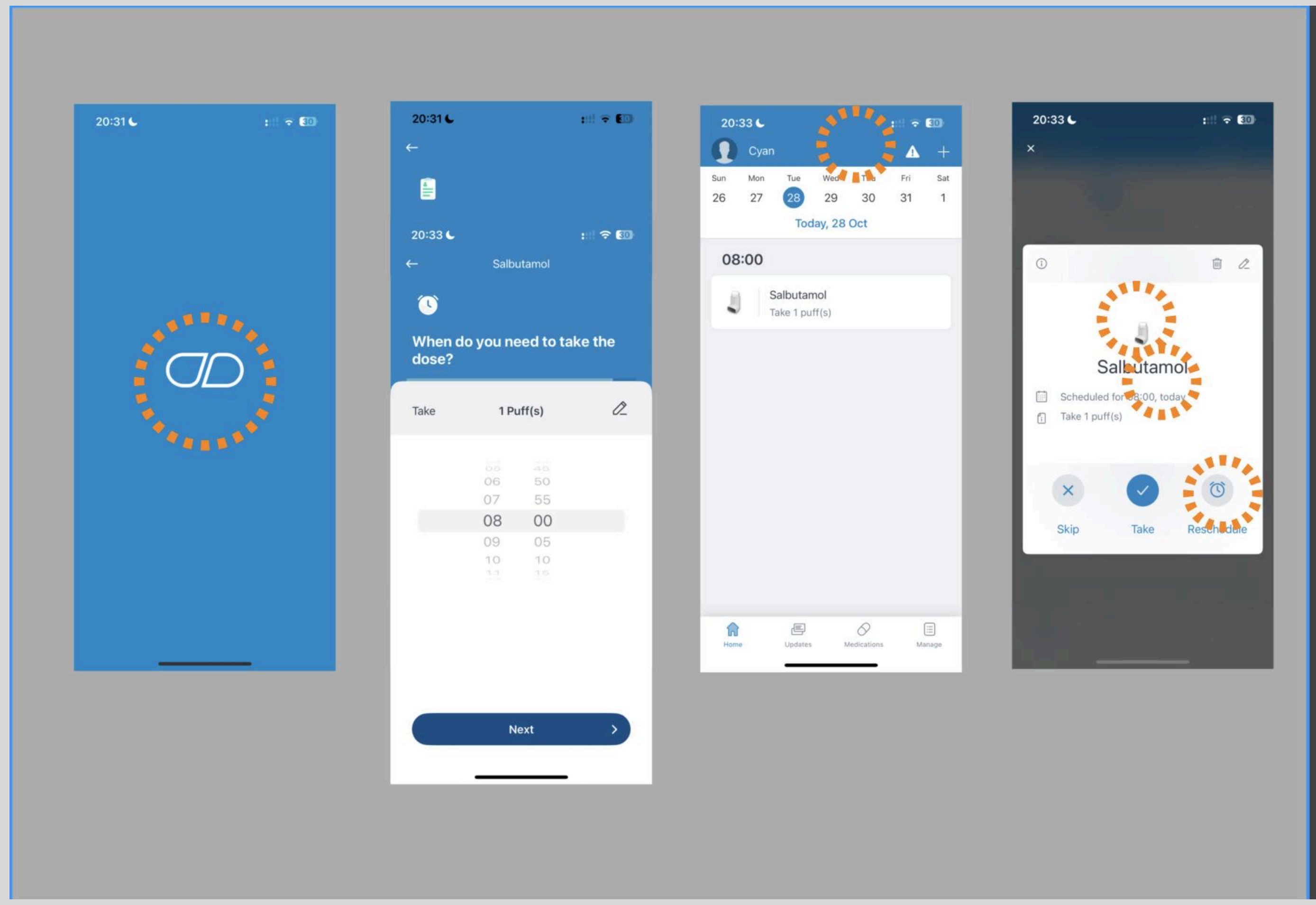
Other nutrition/medication tracker apps

Simple, flat designs are used to offload the user's cognition by projection. Simple colour schemes keep the user interface clean. Some icons used have weak connections to what they represent, making some of the apps affordances unclear

Moving on to Broker 2 and focusing on signifiers and affordances, my team designed a nutrition tracking app for older adults(65+), as many of the apps out there have focuses on weight loss or another specific cause, and many are not designed with the demographic in mind.

We designed a mobile application that can help users track their food consumption, nutrition levels, and suggest healthier and easier meal alternatives to support everyday cooking and lifestyle habits, in a way that is accessible to people with these navigation challenges.

The intended demographics and PACT(People, Activity, Context, and Technology) framework of our designs became a large concern that I had not thought about for Broker 1.



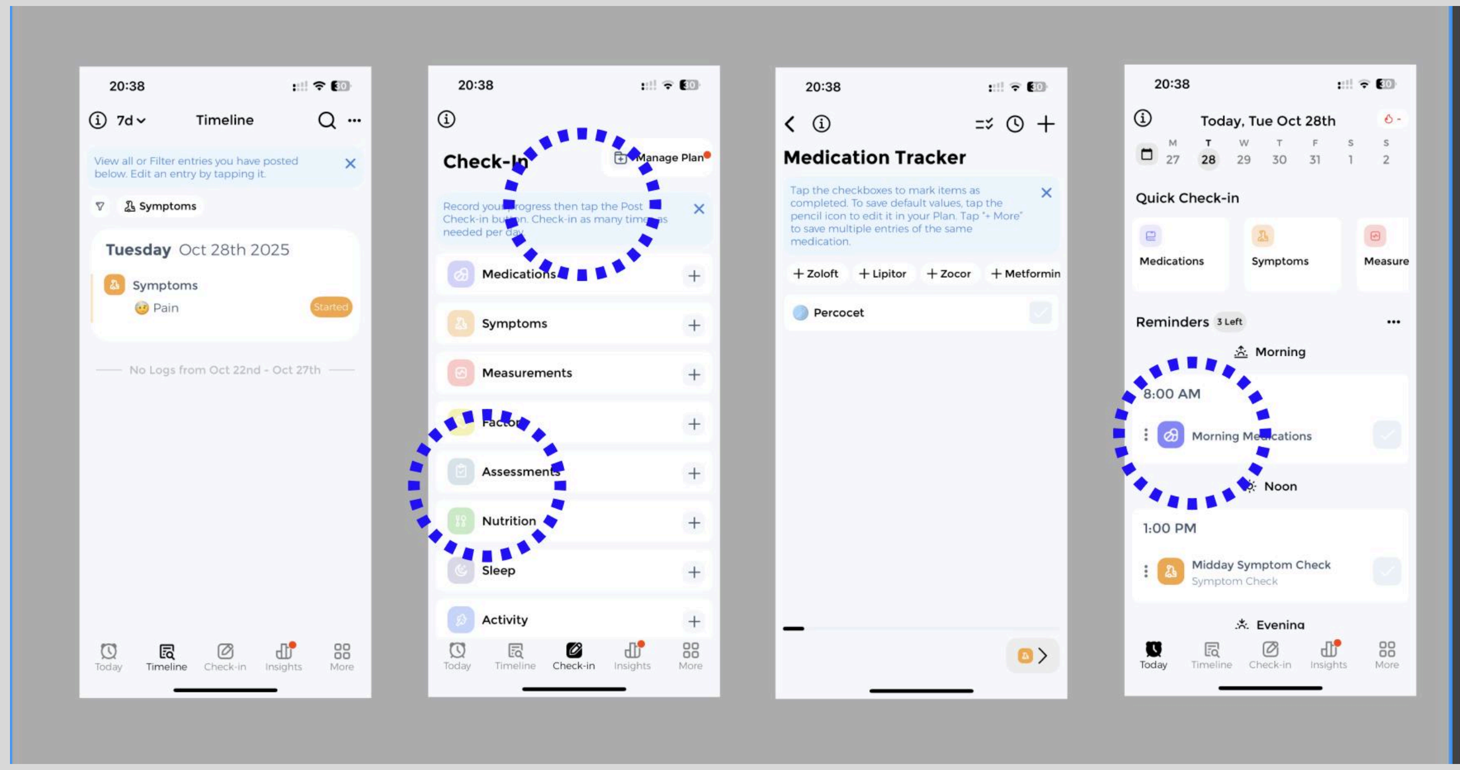
Medisafe

Flat design interface (e.g. clock) → Clear visual hierarchy and low cognitive load.

Skeuomorphic (medicine) strengthens perceived affordance.

Consistent layout and repetition construct users' habit and reduce work memory load.

External cognition augment intelligence.



CareClinic

Flat design with emoji, weaker perceived affordance.

Lack of contrast makes it harder for older users to distinguish the elements, increase cognitive load.

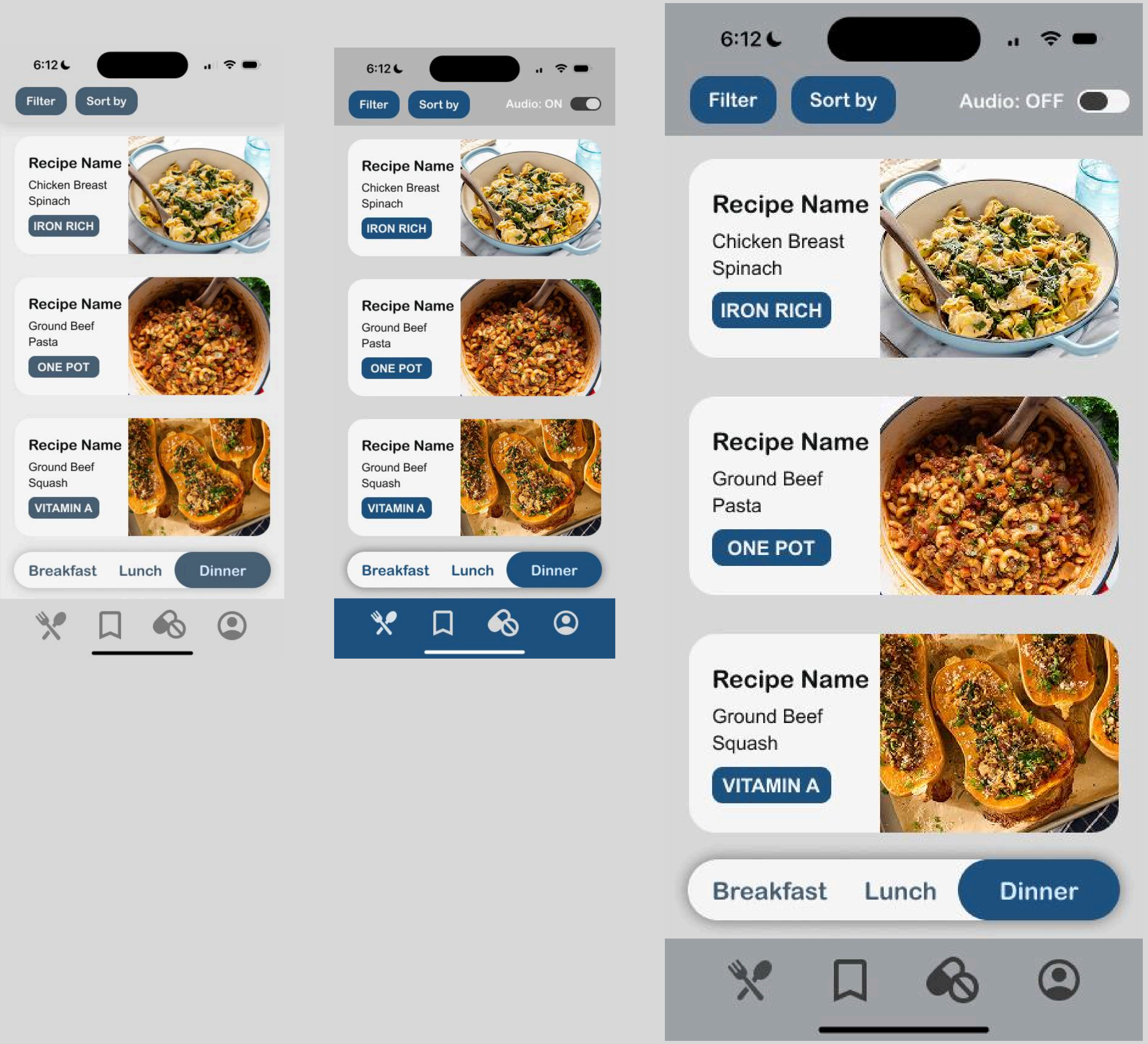
Design iterations(L) and final outcome(R)

Our group reflected on our designs from a variety of perspectives and iterated based on feedback, increasing the contrast and the vibrance to make it easier to use and more appealing for older demographics

I learned a lot about creating clear signifiers and affordances while creating this design: we wanted the app to be simple and easy to use by ensuring UI icons were not too small and were simple to understand and meaningful.

Alternative UI elements, ("Filter" and "Sort by" buttons, slider for audio assistance, Breakfast/Lunch/Dinner selection) are all inspired by similar design elements on other familiar apps, tapping into the users memory to make the design feel easy to use.

We used smart-seeing and the gestalt law of proximity to group the recipes in cards and the interfaces different coloured containers, helping the user to know that items inside are related - icons at the bottom all take you to different pages while buttons at the top help navigate the current page, etc.



Modified design, saved recipes page

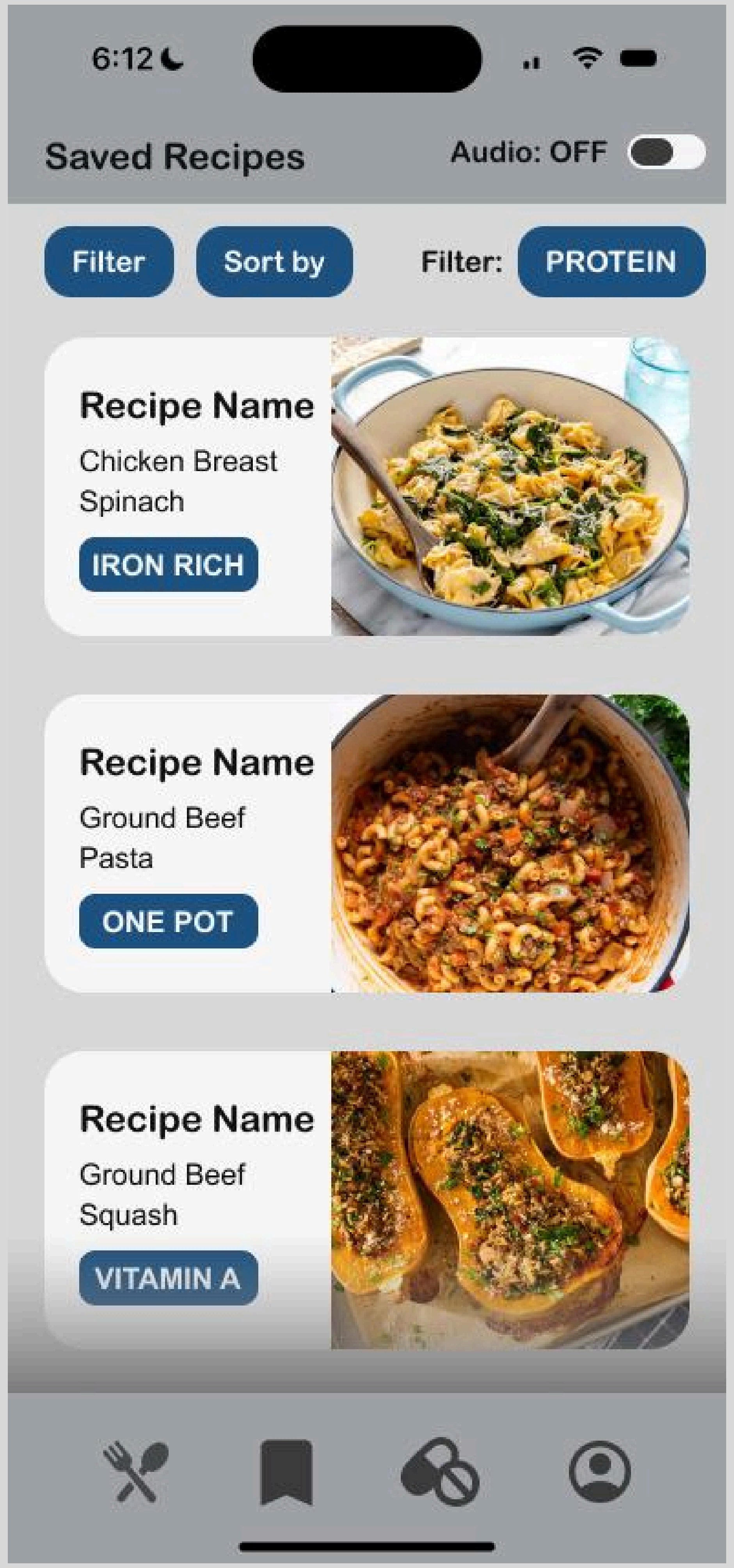
Clear title, modified filtering, clear signifier of current page.

After receiving feedback and reflecting on our design, I have made a few minor tweaks to make it clearer for the user:

I created a mockup of what the “Saved Recipes” bookmark menu will look like: this menu helps the user project their consciousness by saving information they can come back to at a later date, such as when they actually want to make the meals. I have modified the filtering interface to reflect this while retaining a clear information hierarchy.

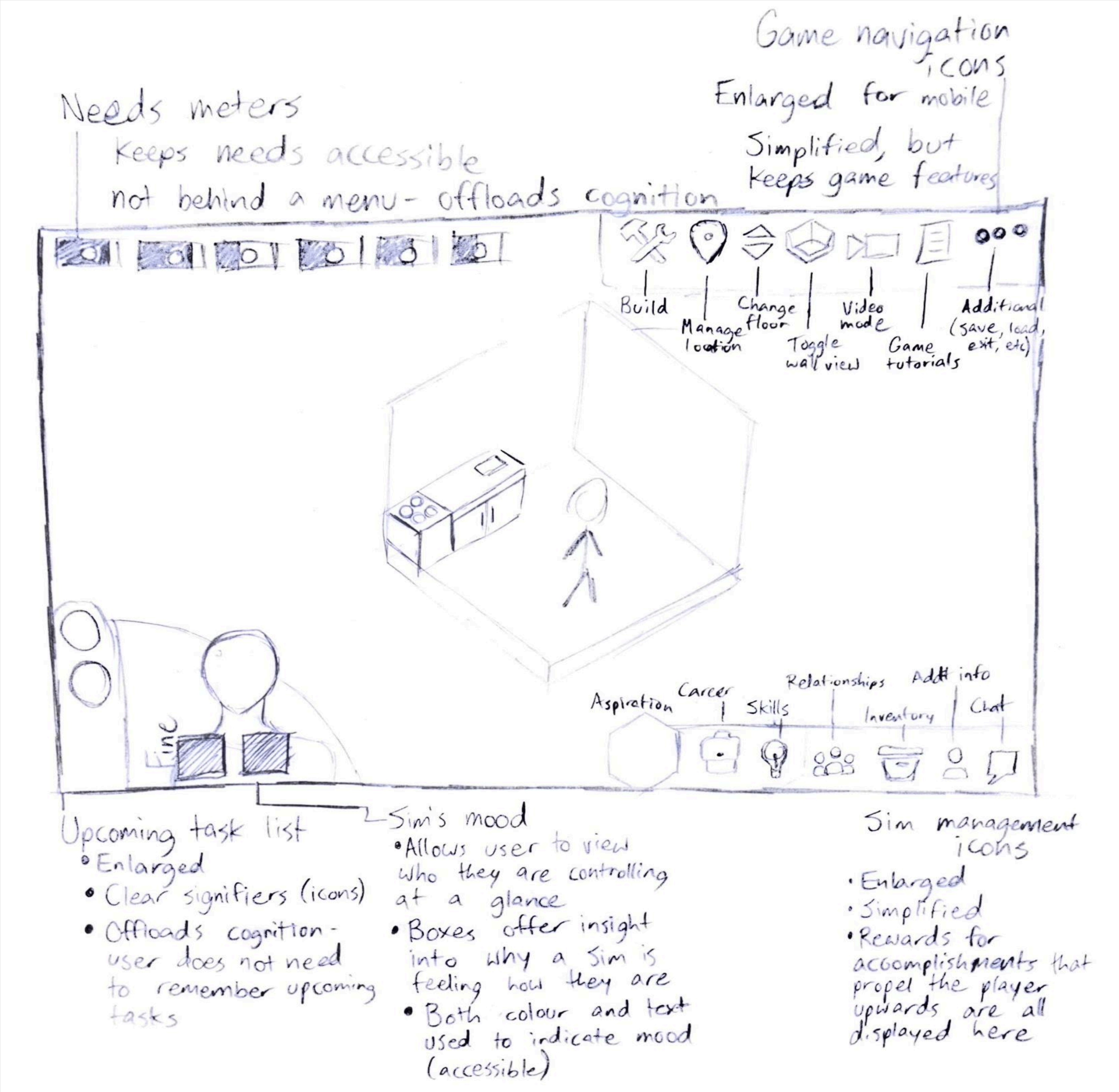
The icons at the bottom will now fill/get thicker when currently on that page. This clarifies the signifier and lessens the chance that a user will try to click on an icon for the page they are already on and get frustrated when nothing happens. I have also increased the contrast of the “Audio” slider at the top of the screen.

Overall, these changes improve the distributed cognition of the design so that the app can be used with minimal friction.



Sketch for an improved game interface

Smart seeing utilized to group icons by category in each corner of the screen. Offloads the players cognition by keeping track of important information such as the Sims' needs and upcoming tasks(previously, needs were hidden behind menus in the bottom right).



For Broker 3, focused on distributed cognition, my group and I played The Sims 4, a life simulation game in which the user experiences life through a virtual "Sim," and we discovered a few issues with the current game:

- It throws you into the most complex customizable interfaces right away
- It has a lot of complexity, which is good for experienced players but is overwhelming for new players.
- The game needs more visual + audio cues for what is happening, including clearer ui elements + signifiers

The immediate exposure to complex UIs and features, as well as the lack of clear signifiers and affordances, overloads the users cognition, which we set out to improve using a mix of sound cues, movement, grouping, and clear, meaningful icons. Our research focused on how to create a clear, fun tutorial, and how to keep the complexities of a UI system while lowering the friction for new players: this was essential to advance us through our process.

Paper prototype of improved game

Follows smart-seeing, projection, and distributed cognition ideation from our paper sketch. The user interface is enlarged and simpler than the Sims 4, but experienced users can still find any features they need. This interface makes things much simpler for new users to jump in and enjoy the game faster.

Our new game design improves the gameplay environment to make it work for the player rather than against them.
Icons are grouped more clearly than in the original game, and the improved design uses smart seeing, projection, and close coupling to help the user move through the game effortlessly

The game holds data such as a Sim's wants and needs and breaks it down into small steps for the user (for example, “this Sim is upset because they have the Active trait and have not exercised in a while”). The user can take simple steps to address this, and never has to internalize what a Sim may be wanting.

The game uses systems such as a “point and click to move”, as well as “needs” bars, which are both commonly used in other games(for example, League of Legends, and Minecraft, respectively) Because so many games use these features, the game mechanics do not feel unfamiliar to try for the first time.

Current Character Creator vs Refined Design

Character screen refined to help user use smart seeing and close-coupling - overwhelming icons(top picture) are simplified and moved to outfits, and premade outfits are placed in a more isolated, distinct zone, with a tutorial popup explaining that they are an option.

In our class gameplay sessions, everyone who tried our game was able to accomplish their goals without instruction. Everyone who tested our game agreed that our prototype felt intuitive, and the mechanic of clicking items to interact with them used the players existing memory of other apps and games to make things easy to adjust to.

In the future, I would focus on ways to simplify the game's existing complex menus in addition to shuffling when they first appear. In the character customizer, for example, the series of icons displaying different garments and accessories is overwhelming(19 icons!) and unclear at such a small scale. They do not follow the Gestalt law of similarity: the inner circle of icons is mostly for the types of clothing(ie pants, shirt, shoes), but there is also a button for full premade outfits hidden in there. This should reside in a separate place on the screen where it is more clear that the user can bypass the game's HUNDREDS of clothing options.

I have created a prototype to reflect this, simplifying the menu for a user so it is less overwhelming and works for them, rather than against them.

Throughout these Broker assignments, I have learned and solidified key concepts of human-centred design such as smart-seeing, projection, close-coupling, and how I can best make the user's environment support them. I have learned how to make affordances clear using meaningful signifiers and how to implement them.

I have learned a lot about accessibility in design and seen the importance of checking my designs for colour blindness, motor difficulties, or any other accessibility concerns that may apply to my intended graphic.

I have realized the importance of receiving feedback and reflecting and iterating on my designs accordingly, and experienced how much it can make or break designs.

I will be taking all of these ideas into any future designs I may create, bringing a fresh perspective to designing as a whole.

Thank you.
Norah Kingsland